Stochastic Simulation assignment 1A

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1 Problem description

We want to simulate an electric vehicle (EV) charging station with $c = 4$ identical charging points (servers). Vehicles arrive throughout the day, join a shared queue if all chargers are busy, and may abandon the queue if the wait is too long. Each vehicle charges for a time that depends on its battery state at arrival. At time 0 all chargers are idle and the queue is empty.

1.1 Vehicle Arrival Process, Non-Stationary

The inter-arrival time between vehicle n and vehicle $n+1$ (in minutes) is: $\Delta t_n = 15 \cdot (1 + \sin(n\pi/12))^2 + 2$. This deterministic formula produces a non-stationary arrival pattern with busy and quiet periods, mimicking the variation of EV demand throughout a day.

1.2 Service Time, Battery-Dependent

Each vehicle n arrives with a battery level $b_n \in (0, 1) : b_n = 0.5|\sin(n\pi/7) + 1|$. The service time (charging duration in minutes) required to reach a full battery is: $S_n = 60 \cdot (1 - b_n)$.

1.3 Reneging Mechanism (new event)

A vehicle that joins the queue will renege (leave without charging) if its waiting time exceeds its individual patience threshold. The patience of vehicle n is: $\tau_n = 20 \cdot (1 + |\cos(n \cdot e)|)$ minutes. When vehicle n joins the queue, a reneging event is immediately scheduled for time $t_{arrival,n} + \tau_n$. If a charger becomes free before this event fires, the vehicle begins charging and the reneging event is cancelled. A reneged vehicle is lost and does not return.

1.4 Early Departure (new event)

A vehicle currently at a charger may decide to leave early if it observes a long queue. Specifically, when a new vehicle joins the queue and the queue length reaches a multiple of 5 (i.e. $5, 10, 15, \ldots$), each vehicle currently charging with remaining time above 15 minutes independently interrupts its session with probability 0.2. If it interrupts, an early departure event is scheduled 2 minutes later (the vehicle wraps up and leaves), freeing the charger earlier than planned. The original end-of-service event for that vehicle is cancelled.

1.5 Simulation Horizon

Let T be the time at which the 800th vehicle completes its charging session (early or full). Run the simulation until time T.

2 Problem formulation

We want to calculate:

- The average queue length over $[0, T]$.
- The average waiting time of vehicles that successfully started charging (i.e. did not renege).

- The reneging fraction: proportion of vehicles that reneged out of all arrivals up to T.
- The charger utilisation: average fraction of the 4 chargers that are busy over $[0, T]$.
- The early-departure fraction: proportion of completed charging sessions that were early departures.

3 State variables

The state variables are: (some of these might be statistical counters)

- Number of chargers that are busy
- The number of EV's in the queue*
- Number of EV's currently being charged
- Total EV arrivals
- EV's that have completed the charging
- Reneging counter
- Number of EV's that leave early
- Queue length over time
- Number of busy chargers over time
- Sum of waiting times.
- The arrival times of the waiting EV's in the queue*.

4 Events

Events are:

- EV arrivals:
  - EV joins the system with an arrival time, service time, and patience.
  - The next arrival event is scheduled.
  - Either the EV immediately starts charging if a charger is free. Or it joins the queue where a reneging time is scheduled.
- EV departures (fully charged)
  - Number of busy chargers decreases
  - Number of completely charged EV's increases
  - If queue is not empty, the next EV starts charging, and the queue length is decreased by 1
- EV reneging
  - Queue length is decreased by 1
  - Reneged counter is increased
- EV early departure
  - For the EV's currently charging, if their remaining time is bigger then 15. Then with a chance of 20% the charging is canceld, the EV leaves in two minutes and the next EV in the queue starts charging (if the queue is not empty)

5 Metrics

- The average queue length is calculated by the area_queue divided by T. The area_queue is determined by the queue length between two decision moments in the simulation multiplied by the time between the two decision moments.
- The average waiting time is calculated by the total_waiting_time divided by all the cars that started their charging. In start_service we store the wait time of a EV that start charging, and this we add to the total.
- The reneging fraction is calculated as reneged counter divided by arrival counter.
- The charger utilisation is the area_busy divided by 4T. So the total time of all 4 charges over the spend simulation time.
- The early-departure fraction is determined by dividing the early leaving EV's by the number of EV's that completed charging.

6 Flowchart

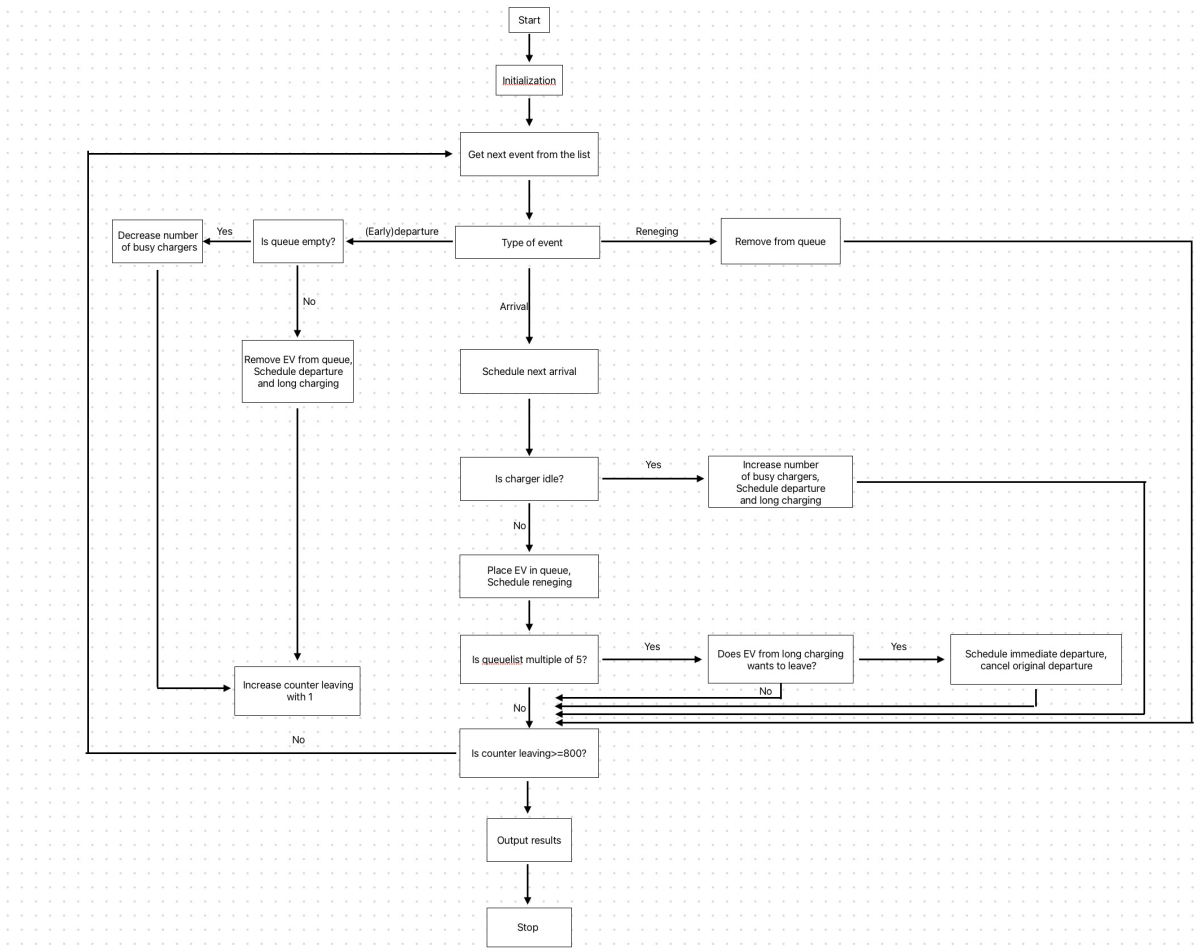

Figure 1: Flowchart of the Electric Vehicle Charging Station model

7 Results and tradeoffs

7.1 Results

- The average queue length over $[0, T]$: 0.27

- The average waiting time of vehicles that successfully started charging: 4.9
- The reneging fraction: 0.068
- The charger utilisation: 0.28
- The early-departure fraction: 0.029

7.2 Trade-offs

In principle we want to minimize the queue length, the amount of people reneging and early departures. When either of these three numbers increases, the other two decrease, direct or indirect.

When people renege, it directly influences the queue length. This also holds for the early departures, because an early departure only occurs when there are people in the queue. By how our model is set up it is clear that changes in the queue length directly influence the amount of early departures.

Early departures are not triggered when the queue length decreases, so reneging does not directly influence the amount of early departures, but it may do so indirectly, because a reneging car can keep the queue length just below being a multiple of 5, resulting in not triggering the early departure. Vice versa, early departures may indirectly influence the amount of reneging people, because the time they win by some other car leaving early may just be the time that makes the difference between a car reneging or not. This is same logic applies for the influence of the queue length on the amount of reneging cars.